

HANIFF HAMDAN

Cloud Engineer | AWS · Terraform · Docker & Kubernetes · CI/CD · Linux

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SUMMARY

Cloud engineer transitioning from three years of full-stack development on government enterprise systems (JPJ, Java/Spring Boot). Built a self-driven cloud portfolio spanning Terraform-provisioned AWS environments (ECS, RDS, OIDC CI/CD), a serverless IoT pipeline, and a human-in-the-loop AI infrastructure agent that diagnoses anomalies with the Gemini API and gates every action behind Telegram approval. *AWS re/Start graduate; AWS Certified Cloud Practitioner exam scheduled.*

CLOUD & INFRASTRUCTURE PROJECTS

AWS Infrastructure Provisioner — Terraform, ECS, RDS, CI/CD | Personal Project

- Provisioned a production-style AWS environment from scratch with Terraform — VPC, ECS Fargate, RDS (MySQL on Graviton/ARM), IAM with GitHub OIDC federation — as a second, independent deployment target for an existing Laravel application alongside its bare-metal Kubernetes (K3s) production environment.
- Built a GitHub Actions CI/CD pipeline authenticating to AWS via OIDC, eliminating long-lived access keys entirely, with automatic Docker builds to ECR and triggered ECS deployments on every push.
- Implemented least-privilege IAM throughout: a scoped Terraform service account, separate ECS execution/task roles, and application secrets pulled at runtime from SSM Parameter Store rather than baked into the container image.
- Diagnosed a multi-stage Docker build failure and restructured the Dockerfile into a two-stage build (Node.js asset compilation feeding a lightweight PHP runtime), resolving a broken frontend asset pipeline.

<http://54.169.236.18/blog/welcome-to-my-blog>

Serverless IoT Smart Plug Pipeline — IoT Core, Lambda, DynamoDB | Personal Project

- Built an edge-to-cloud IoT pipeline: an Orange Pi polls Tuya smart plugs locally via tynytuya, publishes real-time power readings to AWS IoT Core over MQTT/TLS, and triggers a Lambda that enriches each reading against Malaysia's TNB Tariff A before persisting to DynamoDB.
- Designed the pipeline VPC-free with DynamoDB on-demand billing and TTL-based auto-expiry, keeping infrastructure cost near zero for a low-traffic deployment.
- Exposed processed data through a CORS-enabled API Gateway and second Lambda, consumed by a dashboard served via CloudFront with Origin Access Control restricting direct S3 bucket access.
- Resolved cross-environment integration issues spanning Python virtual environment isolation under PM2, module import ordering, and Terraform provider/lockfile conflicts when packaging Lambda functions.

<https://d30t4ra3ip2jtf.cloudfront.net/>

Homelab Infrastructure — Self-Hosted Platform Server | Personal Project

- Operate a self-hosted Kubernetes (K3s) cluster on ARM64 hardware (Orange Pi 4 Pro) running a full observability stack — Prometheus, Grafana, Loki/Promtail — replicating enterprise monitoring and log management workflows with LogQL-based error filtering.
- Built a Gemini 2.5 Flash-powered AI insights layer that continuously reviews system metrics and logs, surfacing plain-language SRE summaries and anomaly explanations pushed to Telegram.
- Designed and deployed HelmDeck, a self-hosted GitOps/Helm deployment manager (NestJS, React/TypeScript, PostgreSQL, BullMQ, Socket.io), framed as an Internal Developer Platform (IDP).
- Independently resolved infrastructure-level networking issues, including K3s iptables chain conflicts blocking inter-container traffic and CSRF failures through a reverse proxy.
- Built and operate a production Laravel application (PHP 8.4-FPM, Nginx, Redis, MariaDB) with repository pattern, soft deletes, DB transactions, and response caching, exposed publicly via a GCP reverse-proxy over Tailscale.

Human-in-the-Loop Infra Monitoring Agent — Gemini, Prometheus, Telegram | Personal Project

- Built a persistent Python agent watching homelab (Prometheus) and IoT (MQTT) metrics, using Gemini 2.5 Flash to diagnose anomalies and propose remediation, with a human approval gate via Telegram inline buttons ensuring no action executes without explicit sign-off.
- Designed a three-layer safety architecture: a harness validating all LLM output before execution, a fixed action allowlist (PM2 restart, Docker service restart, K3s scaling clamped to 1–3 replicas), and context engineering supplying a static system profile, JSON-persisted incident history, and live metrics to every prompt call.

- Implemented smart plug idle detection via a local MQTT pipeline for low-latency alerting, with configurable wattage threshold and cooldown logic to prevent alert spam.
- Applied a harness guardrail that downgrades any out-of-allowlist action to a no-op before reaching the executor, keeping the LLM strictly diagnostic even on malformed or hallucinated output.

EXPERIENCE

Full-Stack Developer — Heitech Padu Bhd

09/2023 – Current

Project: JPI Teknologi Digital

- Optimized DB2 database queries supporting a national vehicle registration system, improving processing efficiency and reducing data inconsistencies.
- Developed and maintained backend services in Java (Spring Boot) for vehicle registration and licensing workflows, and built integration tests to validate API reliability.
- Collaborated in Agile sprints, contributing to backlog refinement, sprint planning, and cross-team delivery on a government enterprise system.

Project: KPDRM System Revamp

- Redesigned and modernized legacy system architecture using C# (.NET) and REST APIs, improving SQL query performance and reducing data retrieval time by approximately 30%.
- Led UAT testing cycles, identifying critical issues and reducing production defects by 20%.
- Developed responsive frontend components aligned with backend architecture for improved UX.

System Developer — NIISE, IRIS Corporation Bhd

06/2023 – 09/2023

- Engineered backend APIs in C# and ASP.NET to process and transmit biometric data from fingerprint scanners in real time.
- Integrated hardware authentication systems with web applications for secure verification, and re-engineered immigration system modules to improve scalability.

Software Developer Intern — Bravonet Sdn Bhd

07/2022 – 01/2023

- Developed responsive web interfaces using JavaScript, Bootstrap, and CSS with cross-browser compatibility.
- Supported CI/CD build, test, and deployment workflows alongside senior developers.

SKILLS

Cloud & Infrastructure: AWS (VPC, ECS Fargate, RDS, IAM, IoT Core, Lambda, DynamoDB, S3, CloudFront, API Gateway)

Infrastructure as Code: Terraform (state management, remote backends, modules)

Containers & Orchestration: Docker, Kubernetes (K3s), Helm

CI/CD & Version Control: GitHub Actions (OIDC, environment protection rules), Git

Observability: Prometheus, Grafana, Loki/Promtail, CloudWatch

Linux & Networking: Linux administration, NGINX, MQTT, Tailscale

Databases: MySQL/MariaDB, PostgreSQL, DynamoDB, SQL optimization

Backend Development: Java (Spring Boot), C# (.NET/ASP.NET); also familiar with Node.js, PHP, React

Practices: Agile/Scrum, Test-Driven Development, Microservices Architecture

EDUCATION

Bachelor of Engineering Technology, Electronic (Computer System) — Universiti Malaysia Pahang

2019 – 2023

Diploma, Electronic Engineering (Communication System) — Politeknik Kota Bharu

2015 – 2018

CERTIFICATIONS

- AWS Certified Cloud Practitioner — In Progress (exam scheduled)
- DevOps Foundation Certification

LEADERSHIP & PROGRAM EXPERIENCE

- AIESEC — Director, Outgoing Global Volunteer: Led team operations and stakeholder engagement; managed international partnerships
- PEKA 4 — Discipline & Safety EXCO: Coordinated safety initiatives and compliance awareness programs